Math 257 Definitions and Theorems

Week 1: Modules 1-4

Fall 2023

1 Introduction to Linear Systems

Definition 1.1. A **linear equation** is a equation of the form

$$a_1x_1 + \ldots + a_nx_n = b$$

where $a_1, ..., a_n, b$ are numbers and $x_1, ..., x_n$ are variables.

Definition 1.2. A **linear system** is a collection of one or more linear equations involving the same set of variables, say, $x_1, x_2, ..., x_n$.

A **solution** of a linear system is a list $(s_1, s_2, ..., s_n)$ of numbers that makes each equation in the system true when the values $s_1, s_2, ..., s_n$ are substituted for $x_1, x_2, ..., x_n$, respectively.

Theorem 1.1. *A linear system has either*

***one unique solution, no solution** or **infinitely many solutions**.*

Definition 1.3. The **solution set** of a linear system is the set of all solutions of the linear system. Two linear systems are **equivalent** if they have the same solution set.

2 Matrices

Definition 2.1. An $m \times n$ **matrix** is a rectangular array of numbers with m rows and n columns.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \qquad a_{ij} \text{ is in the } i\text{th row and } j\text{th column of } A$$

Definition 2.2. Given a linear system,

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

we define its **coefficient** and **augmented** matrices to be

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \text{ and } \left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right], \text{ respectively}$$

Definition 2.3. An **elementary row operation** is one of the following:

- (Replacement) Add a multiple of one row to another row: $R_i \to R_i + cR_j$, where $i \neq j$,
- (Interchange) Interchange two rows: $R_i \leftrightarrow R_j$,
- (Scaling) Multiply all entries in a row by a nonzero constant: $R_i \to cR_i$, where $c \neq 0$.

Remark. Every row operation is reversible.

Definition 2.4. Two matrices are said to be **row equivalent**, if one matrix can be transformed into the other matrix by a sequence of elementary row operations.

Theorem 2.1. *If the augmented matrices of two linear systems are row equivalent, then the two systems have the same solution set.*

3 Echelon Forms of Matrices

Definition 3.1. A matrix is in **echelon form (**or **row echelon form)** if

1. all *nonzero rows* (rows with at least one nonzero element) are above any rows of all zeros, and
2. the *leading entry* (the first nonzero number from the left) of a nonzero row is always strictly to the right of the leading entry of the row above it.

Definition 3.2. A matrix is in **row reduced echelon form** (or: **reduced echelon form**, or **RREF**) if it is in echelon form and the following as satisfied:

3. the leading entry in each nonzero row is 1, and

4. each leading entry is the only nonzero entry in its column.

Theorem 3.1 (Uniqueness of RREF)**.** *Each matrix is row-equivalent to one and only one matrix in reduced echelon form.*

Definition 3.3. We say a matrix B is **the reduced echelon form** (or: the **RREF**) of a matrix A if A and B are row-equivalent and B is in reduced echelon form.

Definition 3.4. A **pivot position** is the position of a leading entry in an echelon form of a matrix. A **pivot column** is a column that contains a pivot position.

Definition 3.5. A **basic variable** (or **pivot variable**) is a variable that corresponds to a pivot column in the coefficient matrix of a linear system. A **free variable** is variable that is *not* a pivot variable.

4 Gaussian Elimination

Goal. Solve linear systems for the pivot variables in terms of the free variables (if any) in the equation.

Algorithm 4.1. (Gaussian Elimination) Given a linear system,

(1) write down the augmented matrix,

(2) find the reduced echelon form of the matrix,

(3) write down the equations corresponding to the reduced echelon form, and

(4) express pivot variables in terms of free variables.

Theorem 4.1. *A linear system is* ***consistent*** *if and only if an echelon form of the augmented matrix has* ***no*** *row of the form* $\left[\begin{array}{ccc|c} 0 & \ldots & 0 & b \end{array}\right]$, *where* b *is nonzero. If a linear system is consistent, then the linear system has*

- *a unique solution (when there are* ***no*** *free variables), or*
- *infinitely many solutions (when there is* ***at least one*** *free variable).*